

Abdur Rahim

Senior AI Engineer | Voice Agents & LLM Infrastructure

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Technical Skills

AI/ML: LLM fine-tuning (LoRA, SFT, DPO, GRPO), vLLM, RAG, HuggingFace, PyTorch, JAX, MLX
Voice AI: Pipecat, LiveKit, Whisper, Parakeet, Ultravox, Kokoro, Deepgram
Agentic: Multi-agent orchestration, LangGraph, ReAct, Function calling, Tool use, MCP
Infrastructure: Python, SQL, Docker, Linux, AWS, CUDA, Multi-GPU serving, Quantization, Vector DBs

Professional Experience

Senior AI Engineer ESAP AI, Saudi Arabia (Remote) May 2025 – Present

- Developed PageSense, a fully on-premises platform that extracts structured, searchable data from messy real-world Arabic business documents, scanned PDFs, phone photos, and handwriting that traditional OCR can't read
- Shipped production voice agents on Pipecat with multi-agent orchestration for enterprise ERP workflows
- Designed multi-model A100 inference stack achieving <500ms TTFB with custom streaming services over WebRTC
- Built LLM service layer integrating Qwen/Llama with RAG retrieval and dynamic tool selection for ERP-domain agents
- Fine-tuned custom Ultravox speech-LLM and Meta's Omnilingual ASR for bilingual Arabic/English ASR
- Delivered Playwright-based agentic web automation with function calling for enterprise operations workflows

Open Source

mlx-tune: LLM/Multimodal Fine-Tuning Framework for Apple Silicon [GitHub \(1.4k+ Stars\)](#)

- Built and maintain an open-source MLX-native framework with an Unsloth-compatible API for LLM/VLM/Audio/OCR
- Supports LLMs, vision-language, speech (TTS/STT), embeddings, and OCR models via LoRA/QLoRA across 7 trainer algorithms (SFT, DPO, ORPO, GRPO, KTO, SimPO, CPT)
- Authored pure-MLX CTC/RNN-T/TDT transducer losses; first streaming-ASR training path on Apple Silicon, with LoRA support for 39+ MoE architectures and 50+ end-to-end examples

mlx-dspark: DeepSeek's DSpark Speculative decoding on Apple Silicon [GitHub \(650+ Stars\)](#)

- The MLX port of DeepSeek's DSpark speculative-decoding drafter: lossless **4x** faster Gemma-4/Qwen3/Qwen3.6 inference on Apple Silicon, served via an OpenAI-compatible API with continuous batching and prefix caching.
- Profiled and optimized the speculative-decoding path (cross-attention KV cache, Markov/confidence heads, 4-bit quantization); derived a memory-bandwidth cost model + DFlash/EAGLE3 benchmarks, and fixed a GQA KV-tiling bottleneck, turning a net-negative long-context regression into a flat up to 3x speedup at 12k+ tokens.

kaggle-tpu-lab: open-source LLM inference on free TPUs [GitHub \(500+ Stars\)](#)

- Run frontier-class open LLMs on Kaggle's free TPU v5e-8 as public OpenAI-/Anthropic-compatible endpoints
- GLM-5.3-Flash (320B MoE): first jax/pallas inference engine to run this model on TPU, 64 tok/s decode (262k ctx)
- Qwen3.8-27B (bf16, no quant, native 262k ctx): 130 tok/s single-stream decode and 10,300 tok/s prefill

Competitions

Kaggle Competitions Expert [kaggle.com/rahim3](#) Dec 2019 – Present

- Google Landmark Recognition **Silver Medal** | Google Landmark Retrieval **Silver Medal**
- SIIM-ISIC Melanoma Classification, Top **12%** | OpenVaccine mRNA Degradation, Top **12%**

Education

University of Chittagong Chittagong, Bangladesh Jan 2017 – May 2023
Bachelor of Science in Electrical and Electronic Engineering